

Differentiation

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UAB-IDEA

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Section 1

The derivative

From secant to tangent

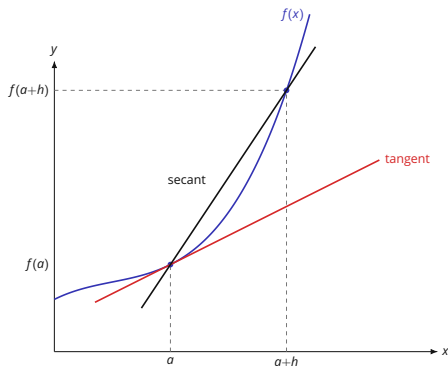
For $f : \mathbb{R} \rightarrow \mathbb{R}$, the slope between a and $b = a + h$ is

$$m_{\text{sec}} = \frac{f(a+h) - f(a)}{h}.$$

Derivative.

$$f'(x_0) = \lim_{h \rightarrow 0} \frac{f(x_0 + h) - f(x_0)}{h},$$

when this limit exists. It is the slope of the tangent at x_0 , obtained as the limit of difference quotients (also called Newton quotients).



As $h \rightarrow 0$, the secant slope converges to the tangent slope.

Video: [Dr. Trefor Bazett, Definition of the Derivative, Part I.](#)

Derivative from the definition

Let $f(x) = x^2$. Start from the difference quotient:

$$\begin{aligned} f'(x) &= \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} \\ &= \lim_{h \rightarrow 0} \frac{(x+h)^2 - x^2}{h} \\ &= \lim_{h \rightarrow 0} \frac{x^2 + 2xh + h^2 - x^2}{h} \\ &= \lim_{h \rightarrow 0} (2x + h) \\ &= 2x. \end{aligned}$$

At $x = 1$,

$$f(1) = 1, \quad f'(1) = 2.$$

The tangent line is

$$\begin{aligned} y - f(1) &= f'(1)(x - 1), \\ y - 1 &= 2(x - 1), \\ y &= 2x - 1. \end{aligned}$$

Derivative from the definition: x^3

A second derivative-from-definition calculation

For $f(x) = x^3$, expand before taking the limit:

$$\begin{aligned}f'(x) &= \lim_{h \rightarrow 0} \frac{(x+h)^3 - x^3}{h} \\&= \lim_{h \rightarrow 0} \frac{3x^2h + 3xh^2 + h^3}{h} \\&= \lim_{h \rightarrow 0} (3x^2 + 3xh + h^2) = 3x^2.\end{aligned}$$

At $x = -2$, $f(-2) = -8$ and $f'(-2) = 12$, so the tangent is

$$y + 8 = 12(x + 2).$$

The function value gives the height of the graph; the derivative gives its local rate of change.

Differentiability & continuity

For $f : \mathbb{R} \rightarrow \mathbb{R}$, differentiability at x_0 means that the derivative limit exists. Equivalently,

$$f(x_0 + h) = f(x_0) + f'(x_0)h + r(h), \quad \frac{r(h)}{|h|} \rightarrow 0.$$

The derivative is the coefficient of the **best local linear approximation**; geometrically, it is the tangent slope.

- ◆ Differentiable at $x_0 \Rightarrow$ continuous at x_0 .
- ◆ A continuous function can have a kink, as the next example shows.
- ◆ If f' is continuous on its domain, f is **continuously differentiable** (C^1).

The same local-linear formula reappears in several variables as $F(x + h) = F(x) + DF(x)h + o(\|h\|)$. In one variable, the linear map $DF(x)$ is simply multiplication by $f'(x)$.

Video: [Khan Academy](#), *Proof: differentiability implies continuity*.

Why differentiability implies continuity

For $x \neq x_0$, factor the change in the function as

$$f(x) - f(x_0) = \frac{f(x) - f(x_0)}{x - x_0} (x - x_0).$$

If f is differentiable at x_0 , then, as $x \rightarrow x_0$,

$$\frac{f(x) - f(x_0)}{x - x_0} \rightarrow f'(x_0), \quad x - x_0 \rightarrow 0.$$

Therefore $f(x) - f(x_0) \rightarrow 0$, so $f(x) \rightarrow f(x_0)$.

The remainder form gives the same conclusion: $f(x_0 + h) - f(x_0) = f'(x_0)h + r(h) \rightarrow 0$ because $r(h) = o(|h|)$.

Continuous need not mean differentiable

Let $f(x) = |x|$. It is continuous at zero because

$$\lim_{x \rightarrow 0} |x| = 0 = f(0).$$

Right derivative.

$$\begin{aligned} f'_+(0) &= \lim_{h \rightarrow 0^+} \frac{|h| - |0|}{h} \\ &= \lim_{h \rightarrow 0^+} \frac{h}{h} = 1. \end{aligned}$$

Left derivative.

$$\begin{aligned} f'_-(0) &= \lim_{h \rightarrow 0^-} \frac{|h| - |0|}{h} \\ &= \lim_{h \rightarrow 0^-} \frac{-h}{h} = -1. \end{aligned}$$

Since $f'_-(0) \neq f'_+(0)$, the derivative $f'(0)$ does not exist.

Section 2

Rules and local approximation

Elementary derivatives

Let the functions inside each formula be differentiable where the formula is applied.

Constant $(c)' = 0.$

Power $(ax^n)' = anx^{n-1}$; for real powers, work on the domain where the expression is defined.

Exponential $(e^g)' = g'e^g, \quad (a^g)' = g' \ln(a) a^g, \quad a > 0, a \neq 1.$

Logarithm $(\ln g)' = g'/g, \quad (\log_a g)' = g'/((\ln a)g), \quad g > 0, a > 0, a \neq 1.$

Trigonometric $(\sin x)' = \cos x, \quad (\cos x)' = -\sin x, \quad (\tan x)' = \sec^2 x.$

Video: 3Blue1Brown, *Derivative formulas through geometry*.

Elementary rules: worked example

Let

$$f(x) = 3x^4 - \frac{2}{x} + e^x - \ln x, \quad x > 0.$$

Rewrite the reciprocal term as $2x^{-1}$, then differentiate term by term:

$$\begin{aligned} f'(x) &= (3x^4)' - (2x^{-1})' + (e^x)' - (\ln x)' \\ &= 12x^3 + 2x^{-2} + e^x - \frac{1}{x} \\ &= 12x^3 + \frac{2}{x^2} + e^x - \frac{1}{x}. \end{aligned}$$

The condition $x > 0$ is required by $\ln x$.

Combining functions

Sum

$$(g + h)' = g' + h'.$$

Product

$$(gh)' = g'h + gh'.$$

Quotient

$$\left(\frac{g}{h}\right)' = \frac{g'h - gh'}{h^2}, \quad h \neq 0.$$

Chain

$$(g \circ h)'(x) = g'(h(x))h'(x).$$

Product and chain: worked examples

Product rule. Let $F(x) = (x^2 + 1)e^x$. Then

$$\begin{aligned}F'(x) &= (x^2 + 1)'e^x + (x^2 + 1)(e^x)' \\&= 2xe^x + (x^2 + 1)e^x \\&= e^x(x + 1)^2.\end{aligned}$$

Chain rule. Let $G(x) = \ln(1 + x^2)$. With $u(x) = 1 + x^2$,

$$\begin{aligned}G'(x) &= \frac{1}{u(x)} u'(x) \\&= \frac{1}{1 + x^2} \cdot 2x \\&= \frac{2x}{1 + x^2}.\end{aligned}$$

Derivative rules

Differentiate each function on its stated domain.

1. $f(x) = 5x^4 - 3x^{-2} + \sqrt{x}, x > 0.$

2. $g(x) = e^{2x} + \ln(3x), x > 0.$

3. $h(x) = x^2 \sin x.$

4. $q(x) = \frac{x^2 + 1}{x - 1}, x \neq 1.$

5. $r(x) = \cos(x^3).$

6. $s(x) = \ln(1 + e^x).$

Identify the outermost operation first, then apply the corresponding rule.

Derivative rules

$$f'(x) = 20x^3 + 6x^{-3} + \frac{1}{2\sqrt{x}},$$

$$g'(x) = 2e^{2x} + \frac{1}{x},$$

$$h'(x) = 2x \sin x + x^2 \cos x.$$

$$q'(x) = \frac{2x(x-1) - (x^2+1)}{(x-1)^2} = \frac{x^2 - 2x - 1}{(x-1)^2},$$

$$r'(x) = -3x^2 \sin(x^3),$$

$$s'(x) = \frac{e^x}{1 + e^x}.$$

Video: 3Blue1Brown, *Visualizing the chain rule and product rule*.

Marginal cost: derivative as rate of change

Marginal cost as a derivative

Suppose

$$C(q) = 100 + 5q + 0.1q^2.$$

Then marginal cost is

$$C'(q) = 5 + 0.2q, \quad C'(20) = 9.$$

The unit is cost per additional unit of output. At $q = 20$, the derivative predicts that one more unit costs about 9.

The actual finite change is

$$\begin{aligned} C(21) - C(20) &= (100 + 5 \cdot 21 + 0.1 \cdot 21^2) \\ &\quad - (100 + 5 \cdot 20 + 0.1 \cdot 20^2) \\ &= 9.1. \end{aligned}$$

Video: [Khan Academy, Marginal cost and differential calculus.](#)

Logarithmic differentiation

A useful trick when a variable appears in the exponent or in a product of powers.

Example 1. $f(x) = x^x, x > 0$.

$$\begin{aligned} y &= x^x, \\ \ln y &= x \ln x, \\ \frac{y'}{y} &= \ln x + 1, \\ y' &= x^x (\ln x + 1). \end{aligned}$$

Example 2. For $A, B, C > 0$ differentiable,

$$\begin{aligned} g &= A^\alpha B^\beta C^\gamma, \\ \frac{g'}{g} &= \alpha \frac{A'}{A} + \beta \frac{B'}{B} + \gamma \frac{C'}{C}. \end{aligned}$$

Video: [The Organic Chemistry Tutor, Introduction to logarithmic differentiation.](#)

Log derivatives, growth rates & elasticities

For a positive differentiable variable $y = y(x) > 0$,

$$\frac{d \ln y}{dx} = \frac{y'(x)}{y(x)}.$$

If $x = t$ is time, this is the instantaneous proportional growth rate $\dot{y}/y$. Thus a small finite change satisfies

$$\Delta \ln y \approx \frac{\Delta y}{y},$$

so log differences approximate proportional changes. If also $x > 0$, the elasticity of y with respect to x is

$$\epsilon_{y,x} = \frac{d \ln y}{d \ln x} = \frac{x}{y} \frac{dy}{dx}.$$

Cobb–Douglas. If $q = AK^\alpha L^\beta$, then

$$d \ln q = \alpha d \ln K + \beta d \ln L.$$

Hence α and β are output elasticities, and $\alpha + \beta$ will reappear as the degree of homogeneity.

Video: [Byron Brown, Logarithms and elasticity](#).

One-variable Taylor polynomials

If f has derivatives through order n at x_0 ,

$$T_n(x) = \sum_{k=0}^n \frac{f^{(k)}(x_0)}{k!} (x - x_0)^k.$$

First order affine approximation

$$T_1(x) = f(x_0) + f'(x_0)(x - x_0).$$

Second order adds local curvature

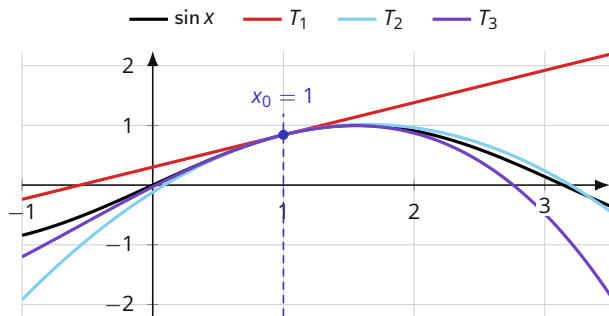
$$T_2(x) = T_1(x) + \frac{1}{2}f''(x_0)(x - x_0)^2.$$

Remainder if $f \in C^{n+1}$ near x_0

$$f(x) = T_n(x) + R_n(x),$$

$$R_n(x) = O(|x - x_0|^{n+1}).$$

Video: [3Blue1Brown](#), *Taylor series*.



Taylor polynomials approximate $\sin x$ near the expansion point $x_0 = 1$.

A quick exercise

Approximate $\ln(1.1)$ using T_1 and T_2 for $f(x) = \ln x$ around $x_0 = 1$.

1 Derivatives at $x_0 = 1$

$$f(1) = 0, \quad f'(1) = 1, \quad f''(1) = -1.$$

3 Evaluate at $x = 1.1$

$$T_1(1.1) = 0.1,$$

$$T_2(1.1) = 0.1 - \frac{0.1^2}{2} = 0.095.$$

2 Taylor polynomials

$$T_1(x) = x - 1,$$

$$T_2(x) = (x - 1) - \frac{1}{2}(x - 1)^2.$$

The actual value is $\ln(1.1) \approx 0.09531$.

The first-order Taylor approximation around the current value of y gives the familiar relation $\Delta \ln y \approx \Delta y / y$.

Section 3

One-variable results

Monotonicity & the Mean Value Theorem

Let S be an interval and let f be differentiable on an open interval containing S .

◆ $f' \geq 0$ on $S \Rightarrow f$ is nondecreasing.

◆ $f' \leq 0$ on $S \Rightarrow f$ is nonincreasing.

◆ $f' > 0$ on $S \Rightarrow f$ is strictly increasing.

◆ $f' < 0$ on $S \Rightarrow f$ is strictly decreasing.

Mean Value Theorem. If f is continuous on $[a, b]$ and differentiable on (a, b) , then there is $c \in (a, b)$ with

$$f(b) - f(a) = f'(c)(b - a).$$

This is the bridge from local slopes to global monotonicity: if $a < b$, then $b - a > 0$, so the sign of every possible $f'(c)$ controls the sign of $f(b) - f(a)$.

Video: [Dr. Trefor Bazett, *The Mean Value Theorem*](#).

The secant slope is attained inside

For $f(x) = 4x - x^2$ on $[0, 3]$,

$$\frac{f(3) - f(0)}{3 - 0} = 1,$$

and $f'(3/2) = 1$.

The secant through the two endpoint values and the tangent at $c = 3/2$ carry the same slope, so the two lines are parallel.

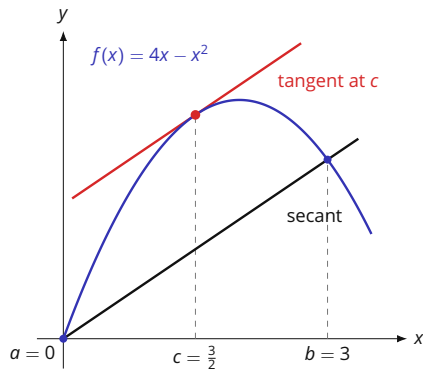


Figure 1 · Secant slope 1, matched by the tangent at $c = 3/2$.

Mean Value Theorem: example

Let $f(x) = x^2$ on $[1, 3]$. The function is continuous on $[1, 3]$ and differentiable on $(1, 3)$.

$$\frac{f(3) - f(1)}{3 - 1} = \frac{9 - 1}{2} = 4.$$

The Mean Value Theorem says that some $c \in (1, 3)$ satisfies

$$f'(c) = 4.$$

Since $f'(x) = 2x$,

$$2c = 4 \quad \Rightarrow \quad c = 2.$$

The theorem's existence claim can be verified directly in this example.

Rolle's theorem

Rolle's theorem. If f is continuous on $[a, b]$, differentiable on (a, b) , and $f(a) = f(b)$, then some $c \in (a, b)$ satisfies $f'(c) = 0$.

Rolle's theorem is the equal-endpoint case of the Mean Value Theorem.

Video: [Krista King, *How to use Rolle's theorem*](#).

Equal endpoint values, horizontal tangent

For $f(x) = x^2 - 4x + 3$ on $[1, 3]$,

$$f(1) = f(3) = 0,$$

so the secant is horizontal, and
 $f'(2) = 0$ at the lowest point
 $f(2) = -1$.

Rolle's theorem is the Mean Value
 Theorem with secant slope 0.

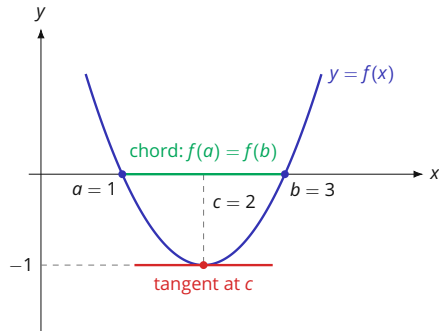


Figure 2 · Chord from $(1, 0)$ to $(3, 0)$, tangent at $c = 2$.

Rolle's theorem: example

Let

$$f(x) = x^2 - 4x + 3 = (x - 1)(x - 3)$$

on $[1, 3]$. Then $f(1) = 0 = f(3)$, and f is continuous and differentiable on the required intervals.

$$f'(x) = 2x - 4.$$

Rolle's theorem predicts some $c \in (1, 3)$ with $f'(c) = 0$. Solving gives

$$2c - 4 = 0 \quad \Rightarrow \quad c = 2.$$

Equal endpoint values force a horizontal tangent somewhere in between.

L'Hôpital's rule

L'Hôpital's rule. Suppose f and g are differentiable on a punctured neighborhood of x_0 , $g'(x) \neq 0$ there, and

$$\lim_{x \rightarrow x_0} \frac{f'(x)}{g'(x)} = L.$$

If either $f(x), g(x) \rightarrow 0$ or $|f(x)|, |g(x)| \rightarrow \infty$, then

$$\lim_{x \rightarrow x_0} \frac{f(x)}{g(x)} = L.$$

One-sided versions use the analogous one-sided hypotheses.

Further reading: [Feldman, Rechnitzer & Yeager, CLP-1, §3.7.](#)

Video: [The Organic Chemistry Tutor, L'Hôpital's rule.](#)

L'Hôpital's rule: example and caution

Indeterminate forms

Use it for a genuine indeterminate form.

$$\lim_{x \rightarrow 0} \frac{e^x - 1}{x} = \lim_{x \rightarrow 0} \frac{e^x}{1} = 1.$$

The original form is $0/0$.

Prefer algebra when it is simpler.

$$\begin{aligned} \lim_{x \rightarrow 1} \frac{x^2 - 1}{x - 1} &= \lim_{x \rightarrow 1} \frac{(x - 1)(x + 1)}{x - 1} \\ &= \lim_{x \rightarrow 1} (x + 1) = 2. \end{aligned}$$

Do not use L'Hôpital automatically.

Section 4

Multivariate derivatives

Partial derivatives & gradient

For $f : \mathbb{R}^n \rightarrow \mathbb{R}$,

$$\frac{\partial f}{\partial x_i}(x) = \lim_{\Delta \rightarrow 0} \frac{f(x_1, \dots, x_i + \Delta, \dots, x_n) - f(x)}{\Delta}.$$

$$\nabla f(x) = (f_{x_1}(x), \dots, f_{x_n}(x))^{\top}.$$

The gradient is the vector of marginal effects.

Example, $y > 0$.

$$f = x^3 + \ln y + 3z^2y:$$

$$\nabla f = \begin{bmatrix} 3x^2 \\ 1/y + 3z^2 \\ 6zy \end{bmatrix}.$$

Video: [Dr. Trefor Bazett, Geometric meaning of the gradient vector.](#)

Differentiability = one local linear map

A map $F : \mathbb{R}^n \rightarrow \mathbb{R}^m$ is **differentiable at** x if there is a linear map $DF(x)$ such that

$$F(x+h) = F(x) + DF(x)h + r(h), \quad \frac{\|r(h)\|}{\|h\|} \rightarrow 0.$$

In coordinates this linear map for the **vector-valued** function F is the **Jacobian**

$$DF(x) = J_F(x) = \left[\frac{\partial F_i}{\partial x_j}(x) \right]_{i \leq m, j \leq n}.$$

Two special cases are

$$f : \mathbb{R}^n \rightarrow \mathbb{R} : \quad Df(x) = \nabla f(x)^\top, \quad f : \mathbb{R} \rightarrow \mathbb{R} : \quad Df(x) = f'(x).$$

If all first partial derivatives of F are continuous in a neighborhood of x , then F is differentiable there.

Caution. Existence of all partial derivatives at a point does not by itself imply differentiability. For $f(x,y) = xy/(x^2 + y^2)$ away from the origin and $f(0,0) = 0$, both partial derivatives at $(0,0)$ are zero, but $f(t,t) = 1/2$; hence f is not even continuous there.

A vector-valued example: the Jacobian at a point

Take $F : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ and the base point $(1, 2)$:

$$F(x, y) = \begin{bmatrix} x^2 + xy \\ y^2 - x \end{bmatrix}, \quad J_F(x, y) = \begin{bmatrix} 2x + y & x \\ -1 & 2y \end{bmatrix}, \quad F(1, 2) = \begin{bmatrix} 3 \\ 3 \end{bmatrix}, \quad J_F(1, 2) = \begin{bmatrix} 4 & 1 \\ -1 & 4 \end{bmatrix}.$$

With the increment $h = (s, s)^\top$, each component is a polynomial in s :

$$F((1, 2) + h) = \begin{bmatrix} 3 + 5s + 2s^2 \\ 3 + 3s + s^2 \end{bmatrix}, \quad F(1, 2) + J_F(1, 2)h = \begin{bmatrix} 3 + 5s \\ 3 + 3s \end{bmatrix}, \quad r(h) = \begin{bmatrix} 2s^2 \\ s^2 \end{bmatrix}.$$

Gradient: Cobb-Douglas example

Let

$$q(K, L) = K^{1/2}L^{1/2}.$$

Holding the other input fixed,

$$q_K(K, L) = \frac{1}{2}K^{-1/2}L^{1/2} = \frac{1}{2}\sqrt{\frac{L}{K}},$$

$$q_L(K, L) = \frac{1}{2}K^{1/2}L^{-1/2} = \frac{1}{2}\sqrt{\frac{K}{L}}.$$

At $(K, L) = (4, 9)$,

$$q_K(4, 9) = \frac{3}{4}, \quad q_L(4, 9) = \frac{1}{3}, \quad \nabla q(4, 9) = \begin{bmatrix} 3/4 \\ 1/3 \end{bmatrix}.$$

Total differential and linear approximation

For a differentiable $f : \mathbb{R}^n \rightarrow \mathbb{R}$,

$$df = \nabla f(x)^\top dx, \quad f(x + \Delta x) \approx f(x) + \nabla f(x)^\top \Delta x.$$

Example. If $f(x, y) = x^2 + xy + 2y^2$, then

$$\nabla f(x, y) = \begin{bmatrix} 2x + y \\ x + 4y \end{bmatrix}, \quad df = (2x + y)dx + (x + 4y)dy.$$

Video: [Dr. Trefor Bazett, How to find the tangent plane: linear approximation.](#)

Directional derivatives & level sets

Let $v \in \mathbb{R}^n$ be a unit vector. The directional derivative of f at x in direction v is

$$D_v f(x) = \lim_{t \rightarrow 0} \frac{f(x + tv) - f(x)}{t}.$$

If f is differentiable at x , then

$$D_v f(x) = \nabla f(x)^\top v.$$

In particular, for the coordinate vector e_i ,

$$D_{e_i} f(x) = \frac{\partial f}{\partial x_i}(x).$$

If a differentiable path $c(t)$ stays on a level set, $f(c(t)) = \bar{f}$, then

$$\nabla f(c(t))^\top c'(t) = 0.$$

At a regular point where $\nabla f(c(t)) \neq 0$, **the gradient is normal to the level set.**

Video: Dr. Trefor Bazett, *Directional Derivatives: What's the Slope in Any Direction?*

By Cauchy-Schwarz,

$$|D_v f(x)| \leq \|\nabla f(x)\|.$$

When $\nabla f(x) \neq 0$, **the steepest local increase** is in the **direction** $\nabla f(x) / \|\nabla f(x)\|$.

Directional derivative: worked example

Function, point, and direction

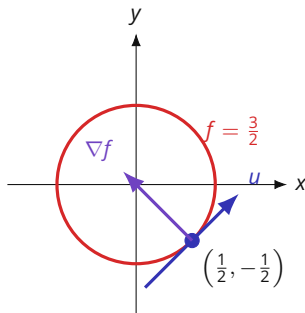
$$f(x, y) = 2 - x^2 - y^2,$$

$$(x_0, y_0) = \left(\frac{1}{2}, -\frac{1}{2}\right), \quad u = \left(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}\right).$$

Calculate with the gradient

$$\nabla f(x, y) = (-2x, -2y), \quad \nabla f(x_0, y_0) = (-1, 1).$$

$$\begin{aligned} D_u f(x_0, y_0) &= \nabla f(x_0, y_0)^\top u \\ &= (-1, 1) \cdot \left(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}\right) = 0. \end{aligned}$$



The direction u is tangent to the level curve, so the first-order change is zero.

Linear approximation: numbers

Continue with $f(x, y) = x^2 + xy + 2y^2$. At $(1, 2)$,

$$\nabla f(1, 2) = \begin{bmatrix} 4 \\ 9 \end{bmatrix}.$$

For $\Delta x = 0.02$ and $\Delta y = -0.01$, the first-order prediction is

$$\begin{aligned} \Delta f &\approx \begin{bmatrix} 4 & 9 \end{bmatrix} \begin{bmatrix} 0.02 \\ -0.01 \end{bmatrix} \\ &= 4(0.02) + 9(-0.01) = -0.01. \end{aligned}$$

The actual change is

$$f(1.02, 1.99) - f(1, 2) = 10.9904 - 11 = -0.0096.$$

Jacobian: local linear map

The Jacobian as the local linear map

Let

$$F(x, y) = \begin{bmatrix} x^2 + y \\ x - y^2 \end{bmatrix}.$$

Then

$$J_F(x, y) = \begin{bmatrix} \partial F_1 / \partial x & \partial F_1 / \partial y \\ \partial F_2 / \partial x & \partial F_2 / \partial y \end{bmatrix} = \begin{bmatrix} 2x & 1 \\ 1 & -2y \end{bmatrix}.$$

At $(1, 1)$,

$$J_F(1, 1) = \begin{bmatrix} 2 & 1 \\ 1 & -2 \end{bmatrix}.$$

If $h = (0.01, -0.02)^\top$, then

$$F((1, 1) + h) - F(1, 1) \approx J_F(1, 1)h = \begin{bmatrix} 0 \\ 0.05 \end{bmatrix}.$$

Video: [Khan Academy](#), *The Jacobian matrix*.

Second derivatives: the Hessian

For a **scalar-valued** function $f : \mathbb{R}^n \rightarrow \mathbb{R}$, differentiate the gradient once more.

Hessian.

$$H_f(x) := J_{\nabla f}(x) = D(\nabla f)(x) = \left[\frac{\partial^2 f}{\partial x_i \partial x_j}(x) \right]_{i,j=1}^n.$$

Column-gradient convention

$$Df(x) = \nabla f(x)^\top, \quad D(\nabla f)(x) = H_f(x).$$

The gradient represents the first derivative of a scalar-valued function; the Hessian is the derivative of that gradient. In one variable these reduce to f' and f'' .

If mixed partial derivatives are continuous in a neighborhood, Young's theorem gives a symmetric Hessian.

Video: [OptWhiz, Gradients, Hessians, and the derivative tests](#).

Hessian of a quadratic function

For $f(x, y) = x^2 + xy + 2y^2$,

$$\nabla f(x, y) = \begin{bmatrix} 2x + y \\ x + 4y \end{bmatrix}, \quad H_f(x, y) = \begin{bmatrix} 2 & 1 \\ 1 & 4 \end{bmatrix}.$$

Gradient and Hessian

For $x > 0$ and $y > 0$, let

$$f(x, y) = 2\sqrt{xy} + x^2 + \ln y.$$

- (a) Compute the gradient $\nabla f(x, y)$.
- (b) Compute the Hessian matrix $H_f(x, y)$.

Gradient and Hessian

GRADIENT

Since $2\sqrt{xy} = 2x^{1/2}y^{1/2}$,

$$\nabla f(x, y) = \begin{bmatrix} \sqrt{y/x} + 2x \\ \sqrt{x/y} + 1/y \end{bmatrix}.$$

HESSIAN

$$H_f(x, y) = \begin{bmatrix} 2 - \frac{\sqrt{y}}{2x^{3/2}} & \frac{1}{2\sqrt{xy}} \\ \frac{1}{2\sqrt{xy}} & -\frac{\sqrt{x}}{2y^{3/2}} - \frac{1}{y^2} \end{bmatrix}.$$

The mixed partials agree, as expected on $\mathbb{R}_{++}^2$.

Second-order approximation

If $f : \mathbb{R}^n \rightarrow \mathbb{R}$ is twice continuously differentiable near x , then as $h \rightarrow 0$,

$$f(x + h) = f(x) + \nabla f(x)^\top h + \frac{1}{2} h^\top H_f(x) h + o(\|h\|^2).$$

Video: [Khan Academy, Vector form of multivariable quadratic approximation.](#)

The quadratic term

The second-order term is a **quadratic form**:

$$Q_x(h) := h^\top H_f(x)h.$$

The Hessian is the symmetric matrix representing this form. The sign of $Q_x(h)$ across directions h connects Taylor approximation to matrix definiteness, convexity or concavity, and second-order optimality conditions.

In one variable, $Q_x(h) = f''(x)h^2$. Thus f' becomes $\nabla f^\top$, while the scalar curvature term becomes $h^\top H_f h$. The remainder $o(\|h\|^2)$ is asymptotically smaller than every fixed multiple of $\|h\|^2$.

The quadratic form in $\mathbb{R}^2$

Let $h = (h_1, h_2)^\top$ and write

$$H_f(x) = \begin{bmatrix} f_{11}(x) & f_{12}(x) \\ f_{21}(x) & f_{22}(x) \end{bmatrix}.$$

Multiplying out the quadratic form gives

$$\begin{aligned} Q_x(h) &= h^\top H_f(x) h \\ &= f_{11}(x)h_1^2 + (f_{12}(x) + f_{21}(x))h_1h_2 + f_{22}(x)h_2^2, \end{aligned}$$

which for continuous mixed partials, where $f_{12}(x) = f_{21}(x)$, becomes

$$Q_x(h) = f_{11}(x)h_1^2 + 2f_{12}(x)h_1h_2 + f_{22}(x)h_2^2, \quad \frac{1}{2}Q_x(h) = \frac{1}{2}f_{11}(x)h_1^2 + f_{12}(x)h_1h_2 + \frac{1}{2}f_{22}(x)h_2^2.$$

The diagonal Hessian entries generate the squared terms; the mixed partial generates the cross term. Some notes absorb the factor $1/2$ into $A = \frac{1}{2}H_f(x)$ and write the Taylor term as $h^\top Ah$. Here we keep the Hessian itself as the representing matrix of $Q_x(h)$, so the Taylor formula contains $\frac{1}{2}Q_x(h)$.

Second-order approximation: numbers

For $f(x, y) = x^2 + xy + 2y^2$,

$$H_f = \begin{bmatrix} 2 & 1 \\ 1 & 4 \end{bmatrix}, \quad h = \begin{bmatrix} 0.02 \\ -0.01 \end{bmatrix}.$$

From the previous linear approximation, the first-order term was -0.01 . The second-order term is

$$\begin{aligned} \frac{1}{2} h^\top H_f h &= \frac{1}{2} \begin{bmatrix} 0.02 & -0.01 \end{bmatrix} \begin{bmatrix} 2 & 1 \\ 1 & 4 \end{bmatrix} \begin{bmatrix} 0.02 \\ -0.01 \end{bmatrix} \\ &= 0.0004. \end{aligned}$$

Hence $\Delta f = -0.01 + 0.0004 = -0.0096$, matching the actual change.

Matrix differentiation: quick reference

We use column gradients and the Jacobian convention $J_F(x) = [\partial F_i / \partial x_j]$. Let $x, a \in \mathbb{R}^k$ and $A \in \mathbb{R}^{k \times k}$.

Linear expressions

$$\frac{\partial(a^\top x)}{\partial x} = a, \quad \frac{\partial(Ax)}{\partial x^\top} = A.$$

Quadratic form: first derivatives

$$\frac{\partial(x^\top Ax)}{\partial x} = (A + A^\top)x, \quad \frac{\partial(x^\top Ax)}{\partial A} = xx^\top.$$

Quadratic form: second derivative.

$$\frac{\partial^2(x^\top Ax)}{\partial x \partial x^\top} = A + A^\top.$$

These identities differentiate the least-squares objective in matrix form.

Video: Prof. Robert Ghrist, *Derivatives of quadratic forms*. Further reading: Boyd and Vandenberghe, matrix calculus appendix.

Matrix differentiation: practice

EXERCISE

Let

$$q(x) = \frac{1}{2}x^\top Ax + b^\top x,$$

$$A = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}, \quad b = \begin{bmatrix} -3 \\ 0 \end{bmatrix}.$$

Find $\nabla q(x)$ and $\nabla^2 q(x)$, then solve $\nabla q(x) = 0$.

SOLUTION

Because $A = A^\top$,

$$\nabla q(x) = Ax + b = \begin{bmatrix} 2x_1 + x_2 - 3 \\ x_1 + 2x_2 \end{bmatrix},$$

$$\nabla^2 q(x) = A.$$

The stationary equations are

$$2x_1 + x_2 = 3, \quad x_1 + 2x_2 = 0,$$

$$\text{so } x^* = (2, -1)^\top.$$

Section 5

Chain rule & implicit differentiation

Chain rule

For differentiable maps $F : \mathbb{R}^n \rightarrow \mathbb{R}^m$ and $G : \mathbb{R}^m \rightarrow \mathbb{R}^\ell$,

$$D(G \circ F)(x) = DG(F(x)) DF(x).$$

When $n = m = \ell = 1$, the matrices are scalars and this becomes the familiar rule

$$(G \circ F)'(x) = G'(F(x))F'(x).$$

Along a path. If $x(t)$ and f are differentiable, then

$$\frac{d}{dt}f(x(t)) = \nabla f(x(t))^\top x'(t).$$

For the straight path $x(t) = x + tv$, evaluating at $t = 0$ gives exactly $D_v f(x) = \nabla f(x)^\top v$.

Chain rule along a path

Let

$$f(x, y) = x^2 + y^2, \quad x(t) = t^2, \quad y(t) = \sin t.$$

For $z(t) = f(x(t), y(t))$, the chain rule gives

$$\begin{aligned} \frac{dz}{dt} &= f_x(x(t), y(t))x'(t) + f_y(x(t), y(t))y'(t) \\ &= 2x(t) \cdot 2t + 2y(t) \cos t \\ &= 4t^3 + 2 \sin t \cos t. \end{aligned}$$

Direct substitution gives $z(t) = t^4 + \sin^2 t$, so the same derivative is obtained.

Video: [Dr. Trefor Bazett, *The Multi-Variable Chain Rule*](#).

Chain rule: derivative ingredients

Let $f(a, b, c) = \ln(ab) + c^2a - 2bc$, and let

$$a = g(x) = \frac{x}{x+1}, \quad b = h(x) = \cos x, \quad c = \ell(x) = e^{2x},$$

on an interval where $g(x)h(x) > 0$.

Outer partial derivatives.

$$f_a(a, b, c) = \frac{1}{a} + c^2,$$

$$f_b(a, b, c) = \frac{1}{b} - 2c,$$

$$f_c(a, b, c) = 2ca - 2b.$$

Inner derivatives.

$$g'(x) = \frac{1}{(x+1)^2},$$

$$h'(x) = -\sin x,$$

$$\ell'(x) = 2e^{2x}.$$

Chain rule: assemble the derivative

For $F(x) = f(g(x), h(x), \ell(x))$, the multivariable chain rule gives

$$F' = f_a(g, h, \ell)g' + f_b(g, h, \ell)h' + f_c(g, h, \ell)\ell'.$$

Substituting the outer partial derivatives and the three inner derivatives,

$$F' = \left(\frac{1}{g} + \ell^2\right) \frac{1}{(x+1)^2} + \left(\frac{1}{h} - 2\ell\right) (-\sin x) \\ + (2\ell g - 2h)2e^{2x}.$$

Implicit differentiation: a level set

Write the circle as the zero level set

$$F(x, y) = x^2 + y^2 - 25 = 0.$$

If locally $y = y(x)$, then differentiating the identity $F(x, y(x)) = 0$ and using the chain rule gives

$$F_x + F_y \frac{dy}{dx} = 0, \quad \frac{dy}{dx} = -\frac{F_x}{F_y} = -\frac{x}{y}, \quad F_y \neq 0.$$

This is the same geometry as the earlier level-set result:

$$\nabla F(x, y)^\top \begin{bmatrix} 1 \\ dy/dx \end{bmatrix} = 0.$$

Thus the tangent vector is orthogonal to the gradient. At $(3, 4)$,

$$\frac{dy}{dx} = -\frac{3}{4}.$$

Video: [3Blue1Brown, Implicit differentiation, what's going on here?](#)

Implicit differentiation

Consider

$$x^2y^3 + (1+y)^x = x + 2, \quad y > -1.$$

Treat the equation as a level set $F(x,y) = 0$. Use implicit differentiation to find

$$\frac{dy}{dx}$$

wherever the derivative is defined, and state where the denominator is nonzero.

Implicit differentiation

Write

$$F(x, y) = x^2 y^3 + (1 + y)^x - x - 2.$$

Then

$$F_x = 2xy^3 + (1 + y)^x \ln(1 + y) - 1,$$

$$F_y = 3x^2 y^2 + x(1 + y)^{x-1}.$$

Since $F_x + F_y dy/dx = 0$,

$$\frac{dy}{dx} = \frac{1 - 2xy^3 - (1 + y)^x \ln(1 + y)}{3x^2 y^2 + x(1 + y)^{x-1}},$$

provided

$$3x^2 y^2 + x(1 + y)^{x-1} \neq 0.$$

Implicit function theorem: one equation

Suppose F is **continuously differentiable** near (x_0, p_0) ,

$$F(x_0, p_0) = 0, \quad F_x(x_0, p_0) \neq 0.$$

Then **locally** there is a **differentiable** function $x = h(p)$ satisfying $F(h(p), p) = 0$, and, with the partial derivatives evaluated at $(h(p), p)$,

$$F_x h'(p) + F_p = 0, \quad h'(p) = \frac{\partial x}{\partial p} = -\frac{F_p}{F_x}.$$

Inverse-function special case. If $y = f(x)$, write $F(x, y) = f(x) - y = 0$. When $f'(x) \neq 0$, the local inverse $x = h(y)$ satisfies

$$h'(y) = \frac{1}{f'(h(y))}.$$

Differentials give the same formula: $F_x dx + F_p dp = 0 \Rightarrow dx/dp = -F_p/F_x$.

Implicit function theorem: a market equilibrium

Let demand depend on a demand shifter a :

$$D(p, a) = a - p, \quad S(p) = p.$$

An equilibrium price $p^*(a)$ satisfies

$$F(p, a) = D(p, a) - S(p) = a - 2p = 0.$$

Since $F_p = -2 \neq 0$, the implicit function theorem gives

$$\frac{dp^*}{da} = -\frac{F_a}{F_p} = -\frac{1}{-2} = \frac{1}{2}.$$

Thus a one-unit outward shift of demand raises the local equilibrium price by one half unit.

Government purchases and equilibrium output

The Keynesian multiplier

Equilibrium output $Y^*(G)$ satisfies

$$F(Y, G) = Y - C(Y - T) - I - G = 0,$$

where taxes T and investment I are fixed. If C is continuously differentiable and $0 < C'(Y^*(G) - T) < 1$, then

$$\frac{dY^*}{dG} = -\frac{F_G}{F_Y} = \frac{1}{1 - C'(Y^*(G) - T)} > 1.$$

Since $C' > 0$, higher equilibrium income induces additional consumption, so the multiplier exceeds one.

Implicit function theorem: systems

System version of the implicit function theorem

Let

$$F : \mathbb{R}^n \times \mathbb{R}^m \rightarrow \mathbb{R}^n, \quad F(x_0, \theta_0) = 0,$$

and suppose F is continuously differentiable near (x_0, θ_0) . If the $n \times n$ Jacobian $D_x F(x_0, \theta_0)$ is nonsingular, then locally the equation $F(x, \theta) = 0$ determines a unique differentiable function $x = h(\theta)$ with $h(\theta_0) = x_0$.

$$D_\theta h(\theta) = -[D_x F(h(\theta), \theta)]^{-1} D_\theta F(h(\theta), \theta).$$

The scalar formula $dx/dp = -F_p/F_x$ is the $n = m = 1$ special case. In economic comparative statics, $F(x, \theta) = 0$ is often the system of first-order or equilibrium conditions.

Generalized least squares

The GLS objective

Let

$$y = X\beta + u, \quad \mathbb{E}[u \mid X] = 0, \quad \text{Var}(u \mid X) = \sigma^2 \Omega,$$

where $y \in \mathbb{R}^n$, $X \in \mathbb{R}^{n \times k}$ has full column rank, and the matrix $\Omega \succ 0$ is known. GLS minimizes

$$Q(\beta) = \frac{1}{2}(y - X\beta)^\top \Omega^{-1}(y - X\beta).$$

Matrix differentiation gives

$$\nabla_\beta Q(\beta) = -X^\top \Omega^{-1}(y - X\beta), \quad \nabla_\beta^2 Q(\beta) = X^\top \Omega^{-1}X \succ 0.$$

Therefore,

$$\hat{\beta}_{\text{GLS}} = (X^\top \Omega^{-1}X)^{-1}X^\top \Omega^{-1}y.$$

When $\Omega = I_n$, this formula reduces to OLS.

Further reading: [Powell, *Aitken's Generalized Least Squares*](#).

Section 6

Homogeneity & homotheticity

Homogeneous functions

Cone. A nonempty set $D \subseteq \mathbb{R}^n$ is a cone if $x \in D$ and $t > 0$ imply $tx \in D$.

Homogeneous of degree k . Let D be a cone. A function $f : D \rightarrow \mathbb{R}$ is homogeneous of degree k if, for all $x \in D$ and $t > 0$,

$$f(tx) = t^k f(x).$$

Topological closedness is a separate property. Euler's theorem below assumes that D is an open cone.

Example on $\mathbb{R}_{++}^2$: $f(x, y) = Ax^\alpha y^\beta$ is homogeneous of degree $\alpha + \beta$.

Further reading: Boyd and Vandenberghe, *Convex Optimization*, §2.1; Bonnissseau, *Optimisation in Euclidean Spaces*, §3.2.7.

An open cone

Fix $x \in D$. Positive scaling traces

$$R_x = \{tx : t > 0\}.$$

Multiplication by $t > 0$ preserves the direction of x and changes only its distance from the origin.

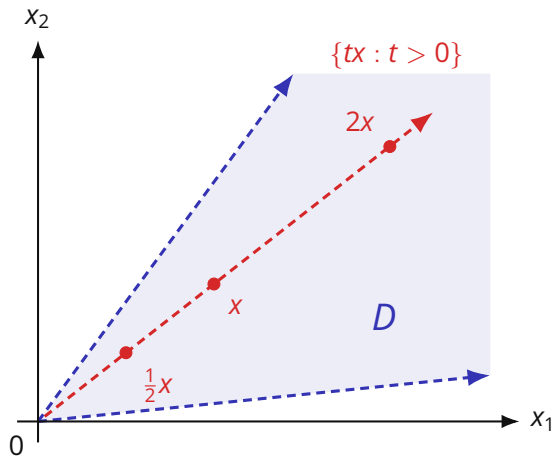


Figure 3 · Positive multiples of x stay in the same cone.

Cobb-Douglas: scaling check

Let

$$f(x, y) = Ax^\alpha y^\beta, \quad A > 0, \quad x, y > 0.$$

For any $t > 0$,

$$\begin{aligned} f(tx, ty) &= A(tx)^\alpha (ty)^\beta \\ &= At^\alpha x^\alpha t^\beta y^\beta \\ &= t^{\alpha+\beta} Ax^\alpha y^\beta \\ &= t^{\alpha+\beta} f(x, y). \end{aligned}$$

Hence Cobb-Douglas is homogeneous of degree $\alpha + \beta$.

Euler's theorem: homogeneity meets the chain rule

Euler's theorem. Let D be an open cone and let $f : D \rightarrow \mathbb{R}$ be differentiable. Then f is homogeneous of degree k iff

$$\nabla f(x)^\top x = \sum_{i=1}^n x_i \frac{\partial f}{\partial x_i}(x) = kf(x).$$

The forward direction comes directly from the chain rule. For fixed x , let $\phi(t) = f(tx)$. Homogeneity gives $\phi(t) = t^k f(x)$, while the chain rule gives

$$\phi'(t) = \nabla f(tx)^\top x.$$

Evaluating both derivatives at $t = 1$ yields $\nabla f(x)^\top x = kf(x)$.

Video: [Mike, the Mathematician, Euler's PDE for homogeneous functions.](#)

Euler's identity in elasticity form

If $f(x) > 0$ and every $x_i > 0$, divide Euler's identity by $f(x)$:

$$\sum_{i=1}^n \frac{x_i}{f(x)} \frac{\partial f}{\partial x_i}(x) = k.$$

Thus the sum of the point elasticities equals the degree of homogeneity. This is the general version of the earlier Cobb–Douglas observation.

Euler's theorem: Cobb-Douglas check

For $f(x, y) = Ax^\alpha y^\beta$,

$$f_x = A\alpha x^{\alpha-1} y^\beta, \quad f_y = A\beta x^\alpha y^{\beta-1}.$$

Therefore

$$\begin{aligned} xf_x + yf_y &= xA\alpha x^{\alpha-1} y^\beta + yA\beta x^\alpha y^{\beta-1} \\ &= \alpha Ax^\alpha y^\beta + \beta Ax^\alpha y^\beta \\ &= (\alpha + \beta)f(x, y). \end{aligned}$$

Thus Euler's identity holds with degree $k = \alpha + \beta$. If $\alpha + \beta = 1$, then $xf_x + yf_y = f(x, y)$.

Homothetic functions: an increasing transformation keeps the level sets

Homothetic. A function $f : \mathbb{R}_{++}^n \rightarrow \mathbb{R}$ is homothetic if $f(x) = h(g(x))$, where g is homogeneous of positive degree and h is strictly increasing on the range of g .

Because h is strictly increasing,

$$f(x) = f(y) \quad \Leftrightarrow \quad g(x) = g(y),$$

so f and g have the same level sets.

Video: [Economics in Many Lessons, Homothetic utility functions and preferences.](#)

Homothetic transformations and level-set slopes

If h is differentiable with $h' > 0$, the chain rule gives

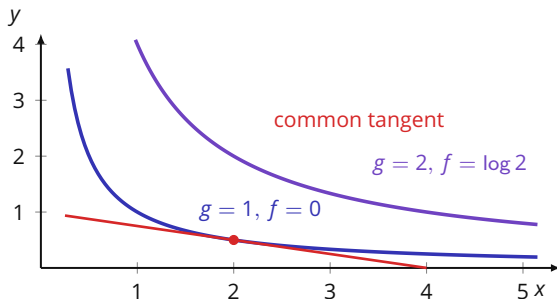
$$\nabla f(x) = h'(g(x))\nabla g(x).$$

Hence whenever the denominator is nonzero,

$$\frac{f_i(x)}{f_j(x)} = \frac{g_i(x)}{g_j(x)}.$$

The common positive factor cancels, so slopes of level sets and MRS are unchanged.

In the figure, $g(x, y) = \sqrt{xy}$ and $f = \log g$. Each curve receives a new level value but keeps the same points and tangent.



A strictly increasing transformation relabels the level curves.

Level-curve slopes are scale-free

If f is differentiable and homogeneous of degree k , and $f_2(x', y') \neq 0$, then for $c > 0$,

$$\begin{aligned} -\frac{f_1(cx', cy')}{f_2(cx', cy')} &= -\frac{c^{k-1}f_1(x', y')}{c^{k-1}f_2(x', y')} \\ &= -\frac{f_1(x', y')}{f_2(x', y')}. \end{aligned}$$

Along a ray from the origin, the marginal rate of substitution is unchanged. A differentiable increasing transformation multiplies both partial derivatives by the same positive factor. The ratio is therefore untouched.

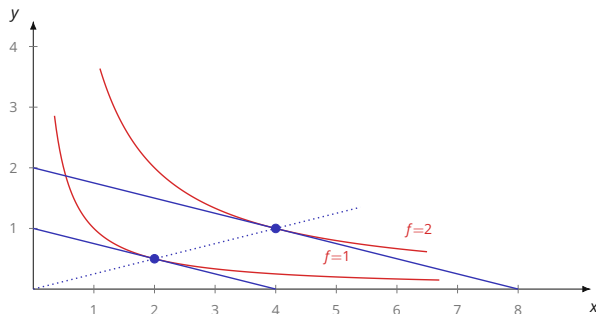


Figure 4 · Isoquants of $f = \sqrt{xy}$; tangents along a ray are parallel.

Section 7

Applications

MRS along an indifference curve

Utility and marginal rates of substitution

An indifference curve is defined by $u(x, y) = \bar{u}$. Total differentiation gives

$$u_x dx + u_y dy = 0, \quad \left. \frac{dy}{dx} \right|_{u=\bar{u}} = -\frac{u_x}{u_y},$$

whenever $u_y \neq 0$. With $MRS_{xy} = u_x/u_y$, the slope is $-MRS_{xy}$. At a regular interior optimum with a binding budget constraint,

$$\frac{u_x}{u_y} = \frac{p_x}{p_y}.$$

Video: [Economicsfun, Calculating the marginal rate of substitution from an indifference curve.](#)

Market-clearing price

Comparative statics

Equilibrium price $p(t)$ solves

$$D(p, t) - S(p, t) = 0,$$

where t is a tax or policy parameter. If $D_p - S_p \neq 0$, the implicit-function formula gives

$$\frac{dp}{dt} = \frac{S_t(p, t) - D_t(p, t)}{D_p(p, t) - S_p(p, t)}.$$

The sign depends on how the policy shifts demand and supply relative to their slopes.

Video: [Prof. Robert Ghrist, *Implicit functions in economics*](#).

Depreciation and steady-state capital

A steady-state comparative static

Let positive steady-state capital $k^*(\delta)$ solve

$$F(k^*, \delta) = f(k^*) - \delta k^* - \bar{c} = 0.$$

If $f'(k^*) - \delta \neq 0$, the implicit-function formula gives

$$\frac{dk^*}{d\delta} = -\frac{F_\delta}{F_k} = \frac{k^*}{f'(k^*) - \delta}.$$

If $f'(k^*) - \delta < 0$, the steady state is locally asymptotically stable. Under this condition, $dk^*/d\delta < 0$: higher depreciation lowers the steady-state capital compatible with the fixed net-output target.

Video: [Mostly Math, Implicit Function Theorem](#).

Estimating equations and GMM linearization

First-order expansion around the population parameter

An exactly identified estimator satisfies $g_N(\hat{\theta}) = 0$, while $g(\theta_0) = 0$. Suppose $\hat{\theta} \xrightarrow{p} \theta_0$, $\sqrt{N}g_N(\theta_0) = O_p(1)$, and

$$\bar{G}_N = \int_0^1 Dg_N(\theta_0 + t(\hat{\theta} - \theta_0)) dt \xrightarrow{p} G = Dg(\theta_0),$$

where G is square and invertible. The mean-value expansion gives

$$0 = g_N(\theta_0) + \bar{G}_N(\hat{\theta} - \theta_0), \quad \sqrt{N}(\hat{\theta} - \theta_0) = -G^{-1}\sqrt{N}g_N(\theta_0) + o_p(1).$$

In overidentified GMM, the sample moment vector is generally nonzero at the estimator; the corresponding estimating equation is the first-order condition for the weighted quadratic criterion.

Arrow-Pratt risk premium

Taylor approximation and risk aversion

Wealth is W and the consumer faces a small fair risk εz , where z is bounded, $\mathbb{E}z = 0$, and $\mathbb{E}z^2 = \sigma^2$. Suppose $u' > 0$ and u is twice continuously differentiable near W . As $\varepsilon \rightarrow 0$,

$$\mathbb{E}[u(W + \varepsilon z)] = u(W) + \frac{1}{2}\varepsilon^2\sigma^2 u''(W) + o(\varepsilon^2).$$

The risk premium π solves $u(W - \pi) = \mathbb{E}[u(W + \varepsilon z)]$ and satisfies

$$\pi = -\frac{u''(W)}{u'(W)} \cdot \frac{\varepsilon^2\sigma^2}{2} + o(\varepsilon^2).$$

Taylor approximation converts local curvature into a monetary measure of risk aversion.

Video: [John Y. Campbell, *The Arrow-Pratt approximation for small risks* \(from 22:08\).](#)